

Q9

A) Computed columns

- a) Computed columns are columns in a table whose values are automatically generated based on an expression or formula involving other columns in the same table. They can be used for various purposes such as generating summary statistics, deriving new columns, or validating data consistency.

B) Table of sines and cosines

- a) $\sin(\theta) + \cos(\theta) = 1$ (sum of $\sin(\theta)$ and $\cos(\theta)$ is equal to 1, then the formula is verified)
 - i) Find the value of $\sin$ for θ degrees, i.e., $\sin(\theta)$.
 - ii) Find the value of $\cos$ for θ degrees, i.e., $\cos(\theta)$.
 - iii) Calculate the sum of $\sin(\theta)$ and $\cos(\theta)$.
 - iv) Compare the result to 1.
- b) Pearson correlation coefficient between columns X and Y
 - i) Calculate the mean of X and Y, denoted by $\bar{X}$ and $\bar{Y}$, respectively.
 - ii) Calculate the standard deviation of X and Y, denoted by S_x and S_y , respectively.
 - iii) Calculate the covariance between X and Y, denoted by $\text{Cov}(X, Y)$.
 - iv) Calculate the Pearson correlation coefficient, denoted by r , using the formula: $r = \text{Cov}(X, Y) / (S_x * S_y)$